

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.IT Semester V)
Applied Business Analytics

Practical III

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```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv("housing.csv")
# Display First Five Records

print(df.head())

# Display Numerical Columns

print(df.select_dtypes(include="number").columns)
```

```
***      longitude  latitude  housing_median_age  total_rooms  total_bedrooms  \
0      -122.23    37.88           41.0           880.0           129.0
1      -122.22    37.86           21.0          7099.0          1106.0
2      -122.24    37.85           52.0          1467.0           190.0
3      -122.25    37.85           52.0          1274.0           235.0
4      -122.25    37.85           52.0          1627.0           280.0

      population  households  median_income  median_house_value  ocean_proximity
0           322.0         126.0         8.3252         452600.0         NEAR BAY
1          2401.0        1138.0         8.3014        358500.0         NEAR BAY
2           496.0         177.0         7.2574        352100.0         NEAR BAY
3           558.0         219.0         5.6431        341300.0         NEAR BAY
4           565.0         259.0         3.8462        342200.0         NEAR BAY
Index(['longitude', 'latitude', 'housing_median_age', 'total_rooms',
      'total_bedrooms', 'population', 'households', 'median_income',
      'median_house_value'],
      dtype='object')
```

```
# Calculate Correlation Matrix

correlation = df.corr(numeric_only=True)
print(correlation)
```

```
***
longitude      longitude  latitude  housing_median_age  total_rooms  \
longitude      1.000000 -0.924664      -0.108197      0.044568
latitude      -0.924664  1.000000      0.011173     -0.036100
housing_median_age -0.108197  0.011173      1.000000     -0.361262
total_rooms      0.044568 -0.036100      -0.361262      1.000000
total_bedrooms    0.069608 -0.066983      -0.320451      0.930380
population        0.099773 -0.108785      -0.296244      0.857126
households        0.055310 -0.071035      -0.302916      0.918484
median_income     -0.015176 -0.079809      -0.119034      0.198050
median_house_value -0.045967 -0.144160      0.105623      0.134153

longitude      total_bedrooms  population  households  median_income  \
longitude      0.069608      0.099773      0.055310     -0.015176
latitude      -0.066983     -0.108785     -0.071035     -0.079809
housing_median_age -0.320451     -0.296244     -0.302916     -0.119034
total_rooms      0.930380      0.857126      0.918484      0.198050
total_bedrooms    1.000000      0.877747      0.979728     -0.007723
population        0.877747      1.000000      0.907222      0.004834
households        0.979728      0.907222      1.000000      0.013033
median_income     -0.007723      0.004834      0.013033      1.000000
median_house_value 0.049686     -0.024650      0.065843      0.688075
```

```

median_house_value
longitude      -0.045967
latitude      -0.144160
housing_median_age 0.105623
total_rooms      0.134153
total_bedrooms    0.049686
population      -0.024650
households       0.065843
median_income     0.688075
median_house_value 1.000000
```

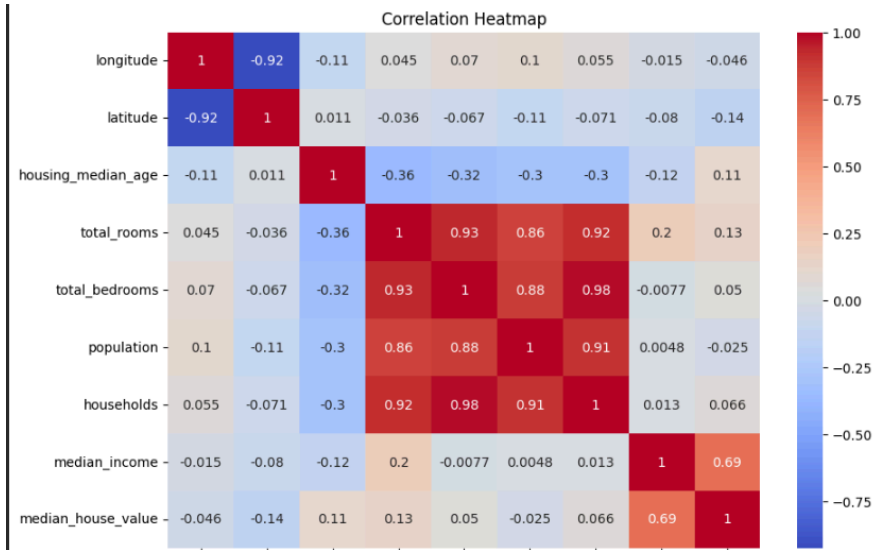
```
# Display Correlation Heatmap

plt.figure(figsize=(10, 7))

sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm"
)

plt.title("Correlation Heatmap")

plt.show()
```



PRACTICAL 3B

```
# Import Libraries

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

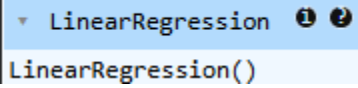
# Load Dataset

df = pd.read_csv("housing.csv")
# Select Independent and Dependent Variables

X = df[["median_income"]]
y = df["median_house_value"]
# Split Dataset into Training and Testing Data

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
# Create Linear Regression Model
```

```
model = LinearRegression()
```

A screenshot of a Jupyter Notebook dropdown menu. The menu is open, showing the selected item 'LinearRegression' with a small downward arrow to its left and two circular icons (one with an 'i' and one with a question mark) to its right. Below the selected item, the text 'LinearRegression()' is visible.

LinearRegression ⓘ ⓘ
LinearRegression()

```
# Display Coefficient and Intercept
```

```
print("Coefficient:", model.coef_[0])
```

```
print("Intercept:", model.intercept_)
```

```
Coefficient: 41933.84939381272  
Intercept: 44459.72916907875
```

```
# Predict House Values
```

```
y_pred = model.predict(X_test)
```

```
# Display First 10 Predictions
```

```
print("Predicted Values:")
```

```
print(y_pred[:10])
```

```
Predicted Values:
```

```
[114958.91676996 150606.88213964 190393.71844449 285059.38345102  
200663.31816103 242165.24890609 257647.22610228 199229.18051176  
245893.1681172 384677.43607096]
```

```
# Calculate Mean Squared Error
```

```
mse = mean_squared_error(y_test, y_pred)
```

```
print("Mean Squared Error:", mse)
```

```
Mean Squared Error: 7091157771.76555
```

```
# Calculate R2 Score
```

```
r2 = r2_score(y_test, y_pred)
```

```
print("R2 Score:", r2)
```

R2 Score: 0.45885918903846656

```
# Scatter Plot with Regression Line

plt.figure(figsize=(8, 5))

sns.regplot(
    x="median_income",
    y="median_house_value",
    data=df
)

plt.xlabel("Median Income")

plt.ylabel("Median House Value")

plt.title("Median Income vs Median House Value")

plt.show()
```

